RxGuard: An Evidence-Grounded Medication Safety Checker

Track A — Clinical Pharmacy & Patient Safety | Week 1: Project Definition & Data Pipeline

1. Clinical Problem and Objective

Medication reviews require clinicians to reconcile multiple drugs, patient-specific risk factors, and lengthy product labels. Inconsistent terminology and alert fatigue can cause important risks to be missed, while ungrounded AI recommendations introduce additional safety concerns. We propose RxGuard, a single-purpose medication safety checker that reviews a pharmacist-entered medication list and essential patient information, flags defined drug–drug interactions (DDIs), contraindications, and selected label-based dose concerns, and links each warning to its source. This four-week project is a proof-of-concept clinical decision support tool, not an autonomous prescribing system.

2. Data Sources and Locked Scope

FDA Drug Labels (primary evidence). Our current curated label_sections.csv contains 775 section records covering 63 drugs and 15 types of label sections, including interactions, contraindications, warnings, and dosing. The records contain drug names, RxCUIs, text, label set identifiers, and DailyMed links. We will audit completeness and relevance to the relevant formulations; openFDA bulk labels provide the source for reproducible updates.

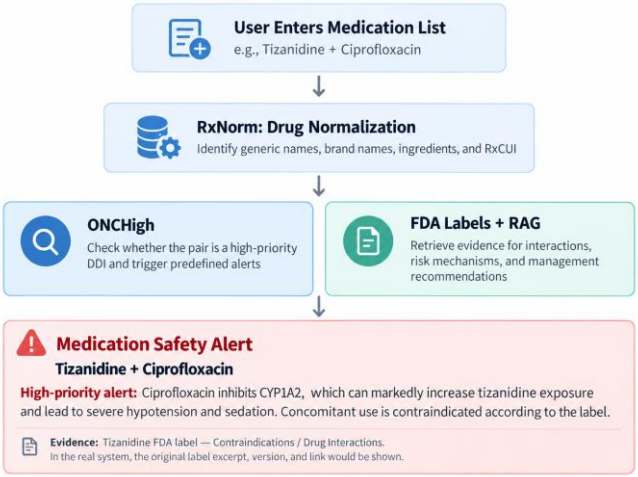
RxNorm (terminology). Normalize brand and generic names to standardized concepts and ingredient identities, maintaining product details when they affect safety rules. Unmatched or ambiguous drugs will be surfaced for review.

ONCHigh (high-priority DDI reference). Use the expert-derived priority interaction list to seed verified DDI rules and design positive tests when drugs overlap our selected formulary. The historical list is nonexhaustive: an unlisted pair must not be declared safe.

Scope lock. RxGuard will have one interface and one output: a medication safety report. Its integrated screening functions are DDI alerts, label-based contraindication alerts when relevant conditions are provided, and a small set of explicitly defined dose warnings when the necessary clinical variables are available. DDI screening is the first development priority. Standalone drug Q&A chatbots, drug summary cards, broad dose optimization, adverse-event prediction, and live EHR integration are out of scope. Synthetic cases will be used for testing; no real patient database is required.

3. Technical Approach

- Input: a pharmacist enters medication names, prescribed doses/routes when available, and relevant patient factors (e.g., age, kidney function, pregnancy status, allergies). Missing items remain unknown, not assumed normal.
- Normalization and retrieval: map names with RxNorm, then retrieve applicable FDA label sections using drug/ingredient filtering and section-aware keyword/vector search (RAG).
- Safety logic: check normalized drugs and patient factors against a curated set of explicit label-supported rules; prioritize verified high-risk interactions and avoid overgeneralizing class-level restrictions.
- Explanation and verification: use an LLM only to phrase evidence-supported alerts. Each output identifies the affected medication(s), safety concern, suggested review action, exact label section, source link, and uncertainty status; unsupported claims are suppressed.



4. Evaluation and Benchmarking

Build approximately 40–60 manually reviewed synthetic test scenarios covering verified high-priority DDIs, contraindication and dosing edge cases supported by selected labels, meaningful non-alert controls, brand-name variants, and missing/ambiguous inputs. Establish expected outcomes from the applicable product labels and independently verify ONCHigh-derived candidate pairs; do not treat absence from ONCHigh as a negative gold standard. Compare the full pipeline with a simple keyword/ingredient-matching baseline.

- Detection: report alert precision, recall, and F1 separately for DDI screening and any other rule categories with sufficient annotated cases; analyze false alerts and misses.
- Terminology: evaluate ingredient-level RxNorm matching accuracy and ambiguity/abstention frequency.
- Citation grounding: measure whether an alert cites the correct drug product and a passage directly supporting its specific risk and action.
- Safety and usability: audit unsupported recommendations, unjustified “safe” claims, handling of insufficient inputs, response time, and completion of typical pharmacist workflows.

Performance estimates will describe this controlled test set only; they will not be presented as validated clinical accuracy or reductions in patient harm.

5. Safety Boundaries

RxGuard will support—not replace—licensed professional judgment. Alerts will display their supporting label text and product provenance; absent or conflicting evidence triggers an explicit uncertainty message. The system will not infer a patient diagnosis, classify off-label use as automatically erroneous, or claim an adverse event has occurred. Dose rules will use the renal metric specified by the chosen label and will not interchange estimated GFR with creatinine clearance. No real patient records will be uploaded or retained for this demonstration.

6. Four-Week Milestones

Week 1 — Data and scope: Finalize the 63-drug formulary; inventory label sections, RxNorm mapping, available ONCHigh pairs, and missing fields; deliver this proposal, an EDA notebook, and a GitHub repository.

Week 2 — Core checks: Implement name normalization, a simple baseline, label retrieval, and a curated DDI ruleset; add a small set of supported contraindication/dose rules if feasible; begin offline tests.

Week 3 — One integrated application: Implement a Streamlit medication-entry form and safety report with exact citations, input-validation safeguards, and abstention behavior; log edge-case failures.

Week 4 — Validation and presentation: Resolve priority error patterns; finalize reproducible code/README, benchmark tables, and a 3–5-minute demonstration video showing a supported high-risk alert and an appropriate uncertainty response.

7. Deliverable and Success Criteria

Deliver a runnable, evidence-grounded medication safety checker—not a general drug chatbot—with reproducible rule/label provenance and transparent performance reporting. Success means reliable identification of the explicitly supported high-risk test cases, verifiable warnings, and safe behavior when evidence or patient details are insufficient.

Key sources: openFDA Drug Labels (open.fda.gov/apis/drug/label/); NLM RxNorm (nlm.nih.gov/research/umls/rxnorm/); Phansalkar et al., JAMIA 2012 (ONCHigh DDI reference).